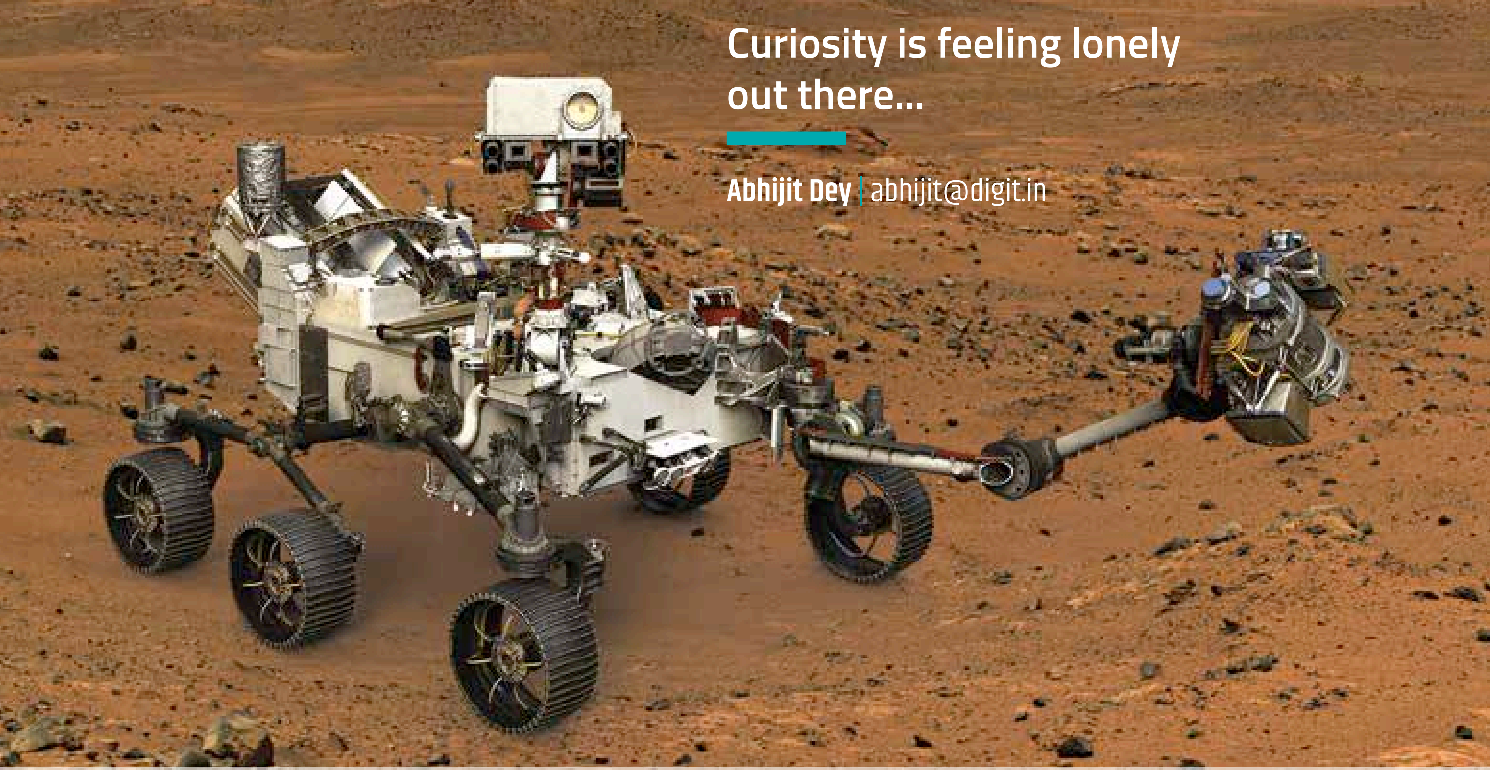
Mars Rover 2020



Curiosity is feeling lonely out there...

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Back in 2013, NASA laid out its grand plan to send a new rover to Mars in 2020. This mission's primary focus is to discover evidence of past microbial life on the Martian planet, collect samples that can eventually be brought back to Earth, and conduct experiments to verify technologies enabling human exploration of Mars in the future. Previous rover missions have already achieved their objectives, with Spirit and Opportunity discovering signs of water on the planet, and Curiosity confirming the presence of environmental conditions in the past that could have supported life on Mars. Now, scientists want to go out and actually look for signs of life on the barren planet. The Mars 2020 rover will introduce new instruments dedicated towards achieving its objectives.

MISSION 2020

The Mars 2020 mission is part of NASA's Mars Exploration Program (MEP). The program consists of four long-term science goals – determining whether life ever existed on Mars, characterising the climate, studying the geology, and finally preparing for human exploration. Most of the goals are based on surface exploration, and although it's highly improbable we will find anything alive on the surface, efforts will be made to search for preserved signs in rock and soil samples.

The first objective will be to find out how the environment used to be and how it changed. The next objective is to search for locations or materials that could have potentially possessed microbial life in the past. Then, if such material is found, the rover will document its findings by storing samples from the site that can be sent back to Earth later. Thus far, we've never brought back any samples from Mars to be studied here on Earth. The Science Definition Team, which is behind outlining the entire mission, has proposed the collection of about 31 samples of rock cores and soil.

The fourth and final objective is to carry out research so that we are fully prepared to send humans to Mars. We hope that once this is accomplished, future missions will not have to worry about certain supplies being sent from earth. One such experiment hopes to be able to scrub oxygen out of the carbon dioxide in the Martian atmosphere, for respiration and also as an oxidiser for